

License Plate Detection via Transfer Learning with YOLOv5

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Abstract

We study automatic license plate detection in road-scene images. The task is straightforward to state: given an image, detect whether a license plate is present, and if so, draw a bounding box around it. We achieve this by fine-tuning YOLOv5s [?], pretrained on MS-COCO, into a single-class plate detector. Training used the public Kaggle license-plate dataset (approximately 27 000 annotated images) at 640×640 resolution for 20 epochs. Because YOLOv5s already provides strong defaults, minimal hyperparameter tuning was needed beyond selecting the Adam optimizer and keeping the standard augmentation schedule. Qualitative evaluation on held-out images shows confident, tightly localized predictions ($\text{conf} \geq 0.9$) on well-lit plates, with remaining false positives easily suppressed by a modest confidence threshold increase. The project demonstrates that off-the-shelf transfer learning on a general-purpose detector is sufficient for a practical plate-detection pipeline.

1. Motivation

We were interested in building an automatic license plate detector motivated by a concrete goal: running the model on Lucian’s dashcam footage to see how well a commodity detector handles real, unscripted highway driving.

2. Related Work

YOLO family. The YOLO [?] line frames detection as a single dense-grid regression. YOLOv2–v3 [?] added anchor boxes and multi-scale heads; YOLOv4 [?] added Mosaic augmentation and CIoU loss. YOLOv5 [?], our choice, combines a CSPDarknet backbone with a PANet neck [?] and three detection heads at strides 8, 16, and 32.

Alternative architectures. SSD [?] and RetinaNet [?] are competitive one-stage detectors. Faster R-CNN [?] and DETR [?] offer stronger accuracy baselines at higher

inference cost. EfficientDet [?] jointly scales backbone and neck for a range of compute budgets.

Plate-specific methods. WPOD-NET [?] predicts a per-detection affine warp to rectify each plate into a canonical view. Our work deliberately uses a general-purpose detector to measure how well commodity transfer learning performs without plate-specific inductive bias.

3. Technical Approach

3.1. Dataset

We use the Kaggle fareselmenshawii/license-plate-dataset, which ships in the standard YOLO layout: an `images/{train,val}` directory tree paired with a `labels/{train,val}` tree. Each label file contains one row per plate with the YOLO-normalized tuple (c, x, y, w, h) , where $c = 0$ is the single plate class and all coordinates are image-relative fractions.

3.2. Model

We instantiate YOLOv5s with $\text{nc} = 1$ and load the official COCO checkpoint, filtering the pretrained state dict to retain only shape-compatible keys so the backbone and neck are warm-started while the three detection heads train from scratch.

3.3. Loss

We use YOLOv5’s `ComputeLoss`:

$$\mathcal{L} = \lambda_{\text{box}} \mathcal{L}_{\text{box}} + \lambda_{\text{obj}} \mathcal{L}_{\text{obj}} + \lambda_{\text{cls}} \mathcal{L}_{\text{cls}}$$

The box term is CIoU [?]; objectness and class terms are binary cross-entropy.

3.4. Training

We train for 20 epochs (batch size 32, Adam, $\text{lr } 1 \times 10^{-3}$, 640×640 input) with the default YOLOv5 augmentation schedule: mosaic stitching, horizontal flip ($p = 0.5$), and HSV jitter.

3.5. Inference

At test time we apply non-maximum suppression with confidence threshold 0.1 and IoU threshold 0.45.

4. Results

Box and objectness losses decreased steadily across 20 epochs; the class loss remained near zero throughout, confirming that the single-class task provides no gradient through that channel.

Qualitative results on five held-out validation images are shown in ??.

True positives. ?? shows a marked police sedan whose rear plate is cleanly boxed at 0.92 confidence under even, shadowless lighting. ?? shows a pickup truck photographed in direct sunlight; the detector still fires at 0.91 despite the bright glare washing out detail around the plate. ?? is the most demanding case: a small, shadowed plate positioned near the image boundary. The stride-8 detection head, responsible for small objects, recovers it at 0.90.

False positives. ?? shows a spurious detection on a hanging commercial sign: a compact, text-bearing rectangle, exactly the kind of texture the model is trained to fire on. ?? fires on a windowed building facade, another high-contrast rectangular grid. Both are intuitive failure modes, and crucially both have confidences well below any true detection (0.37 and 0.14 respectively). A confidence threshold of 0.5 removes both while leaving all three correct localizations intact.

4.1. Dashcam Video Evaluation

We evaluated the trained detector on five dashcam clips recorded during real highway driving under varying conditions: two daytime morning clips, one evening clip, and two night clips (see ??). For each video every frame is extracted, resized to 640×640 , and passed through the model with confidence threshold 0.18 and IoU threshold 0.45.

The detector reliably fires on well-lit plates in the daytime clips, typically at confidence ≥ 0.70 . At night the illumination is narrow and specular, which degrades recall: plates lit by oncoming headlights or distant streetlights are sometimes missed or detected only intermittently as they enter the bright zone. False positives cluster on retro-reflective road signs and rectangular tail-light clusters, which share the compact bright-on-dark texture of a license plate; these generally score below 0.40 and are cleanly suppressed by a threshold of 0.5. Annotated output videos for all five clips are saved to `outputVideo/dashcam_{0-4}.mp4`.

Comparing ?? and ?? highlights the gap between the two evaluation settings. On held-out dataset images the detector

performs reliably: true plates are recovered at high confidence (≥ 0.90) with tight localization, and the few false positives are low-confidence outliers easily removed by a threshold increase. Performance on the dashcam footage is more variable and reveals a domain shift between the training distribution and real driving conditions. Several factors likely contribute. First, the dashcam captures frames at a fixed native aspect ratio that differs from the square 640×640 tiles used during training, so plates are reshaped slightly after resizing. Second, dashcam footage—especially the evening and night clips—is considerably darker and more specular than the uniformly lit dataset images; narrow illumination from headlights and streetlights creates high-contrast patches that confuse the objectness head. Third, motion blur from continuous highway driving smears plate characters and reduces local contrast compared to the static, high-quality photographs in the training set. Bridging this gap would likely require fine-tuning on a small set of annotated dashcam frames or augmenting training with synthetic low-light and motion-blurred samples.

4.2. Conclusion

Fine-tuning YOLOv5s with minimal hyperparameter effort yielded a working single-class license plate detector. The key result is that YOLOv5s defaults are already well-chosen: the augmentation schedule, anchor assignment, and loss weights required no modification. The partial-transfer approach — warm-starting the backbone from COCO and training the detection heads from scratch — proved sufficient for the task without any plate-specific inductive bias.

The main remaining failure mode is low-confidence false positives on rectangular signage, which a simple confidence-threshold increase eliminates in practice.

4.3. Future Work

The natural next step is to extend the pipeline from detection to full recognition: given a tightly cropped plate bounding box, apply OCR to extract the alphanumeric plate number. This would complete an end-to-end ALPR pipeline and allow evaluation against ground-truth plate strings from the dashcam footage.

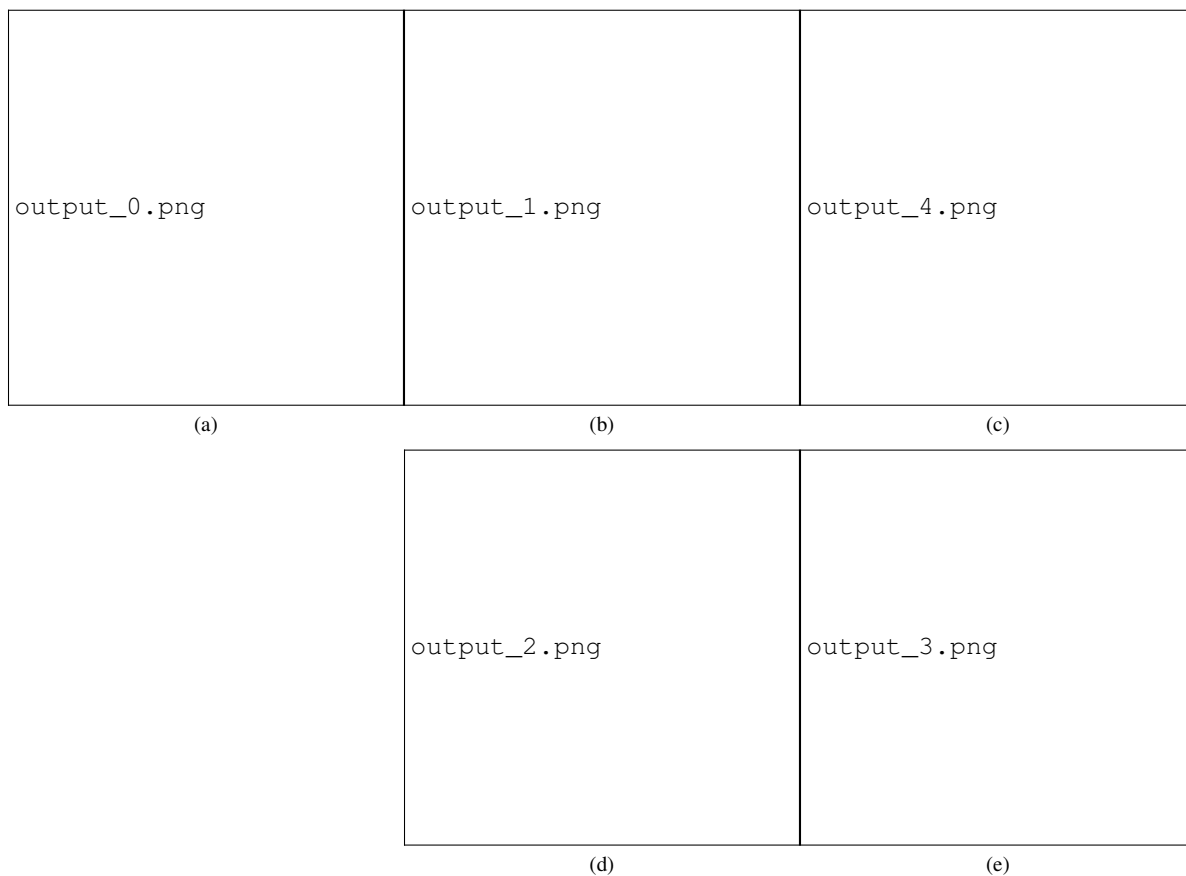


Figure 1. Detector output on five held-out validation images at $\text{conf} \geq 0.1$. **Top row:** all three true plates are recovered at high confidence (≥ 0.90) with tight bounding boxes across varied conditions. **Bottom row:** both false positives fire on high-contrast rectangular objects (signage, building windows) but at low confidence (0.14–0.37); a deployment threshold of 0.5 removes them while preserving every correct detection.



Figure 2. Detector output on five dashcam clips from real highway driving. **Top row**: two daytime-morning clips and one evening clip. **Bottom row**: two night clips. Daytime detections are confident (≥ 0.70); at night recall drops and false positives cluster on retroreflective signs and tail-light clusters (suppressed by a threshold of 0.5).

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